Expressivity Boundaries of Constant-State Attention-Free Sequence Models

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Abstract

Constant-state attention-free sequence models replace content-dependent pairwise scoring with a fixed-size state. We characterize the expressivity boundaries this trade imposes for Spectral Global Convolution (SGC), a causal convolution over a damped-oscillator basis with an FFT parallel form and a second-order IIR streaming form. We prove six theorems: exact real-field duality between the spectral kernel and the two-tap recursion; strict causality; state size independent of sequence length; an equivalence between a Toeplitz impulse response and time-invariant recursion parameters, which entails that context-dependent oscillator parameters destroy the FFT path; an impossibility theorem for content addressing when the history is compressed into a fixed linear summary, valid for arbitrary nonlinear read-outs; and an information-theoretic boundary separating unbounded-length reconstruction from finite-valued task outputs. We then correct the scope of the impossibility theorem: upgrading the input from a vector to an outer-product tensor and the read-out to a bilinear contraction preserves the static LTI kernel, Toeplitz structure, FFT/IIR duality and O(1) state while realizing content addressing. We therefore withdraw the earlier mutual-exclusivity claim between content addressing and FFT duality. On a synthetic single-hop task, the resulting Tensor Spectral Convolution reaches statistical parity with an equal-parameter windowed attention baseline (Fisher exact p = 0.4998) and accuracy 1.0000 with 48.0% of its parameters. All numerical results are finite-sample observations under stated configurations; no measurement is promoted to a theorem. Language modeling at scale, multi-hop retrieval and equal-compute baselines remain unperformed.

1 Introduction

Efficient sequence models with a constant-size inference state are attractive because their memory does not grow with context length. The price is that a fixed state must summarize an unbounded history. The literature has established empirically that such models are weaker than attention on associative recall, copying and state tracking (arXiv:2312.04927, arXiv:2402.01032, arXiv:2404.08819). What has been less precise is *which algebraic property* of a given architecture produces the failure, and whether that property also forbids the FFT/convolution training path. This paper answers both questions for SGC with analytic statements, and then identifies the exact assumption whose removal restores content addressing without reintroducing pairwise scoring.

Throughout, a **theorem** is a universal statement derived only from definitions; a **finite-sample observation** is a measured value tied to a stated configuration, seed count, precision and task; a **conjecture** is an explicitly unverified hypothesis. No measured number is upgraded to a theorem, and no theorem is inferred from a measured number. All observations are reported with their provenance in Appendix D.

The contributions are fourfold, aligned with the theorem groups:

· **C1 (Theorems 1–3).** The spectral kernel and the second-order IIR are the same LTI operator over the reals; the operator is strictly causal; its streaming state is bounded by a constant independent of L. The bitwise fp32 claim is withdrawn and replaced by a path-dependent tolerance statement.

· **C2 (Theorem 4).** Within the two-tap recursion family, a Toeplitz impulse response is equivalent to time-invariant recursion parameters. This strengthens the one-directional non-Toeplitz observation into an equivalence and makes the loss of the FFT path a consequence, not an empirical finding.

· **C3 (Theorems 5–6).** A fixed linear history summary cannot support content addressing for any read-out, including nonlinear ones; a finite state is lossy for unbounded-length reconstruction but can be lossless for finite-valued outputs. Together these separate an architectural defect from an information-theoretic fate.

· **C4 (Section 4).** Tensor signals and a contraction read-out evade the premise of Theorem 5 while preserving every structural property of C1. This yields an explicit constructive counterexample to the earlier mutual-exclusivity claim, plus measured parity and parameter-efficiency results on a synthetic task.

This is a boundary-characterization paper. It does not claim state-of-the-art quality, language-modeling-scale validation, superiority on general language capability, or a formal upper bound on the expressivity of all constant-state models.

2 Preliminaries and Notation

Let x_{1:L} denote a causal token sequence with embeddings u_t in R^N, and let K(τ) be a scalar causal kernel. SGC uses the damped-oscillator family K(τ) = ρ^τ sin((τ+1)θ) with 0 < ρ < 1 and θ in (0, π), extended to M basis functions with distinct (ρ_m, θ_m) and per-channel coefficients. The architecture exposes two evaluation paths: an FFT-based whole-sequence convolution and a streaming two-tap recursion. A *content-addressing* task supplies n key–value pairs in the history and a query q equal to one key; the correct output is the associated value. We write A for the part of the final-layer state that depends on the history but not on q, and B(q) for the part depending on the current query.

The state register is a finite collection of scalar values. Its size is measured in bytes and is independent of L; this is an asymptotic statement and carries a constant factor that grows with the number of channels and basis functions.

Evidence convention. Theorem statements are analytic. Numerical values appear only as finite-sample observations, each accompanied by seed count, precision, task and sample size. Provenance files are listed in Appendix D. Unverified hypotheses are collected in Appendix B and falsified ones in Appendix C.

3 Structural Theorems

Theorem 1 (Exact duality over the reals).
For 0 < ρ < 1 and fixed θ, the convolution y = K * u with K(τ) = ρ^τ sin((τ+1)θ) is identical over the reals to y_t = 2ρ cosθ · y_{t−1} − ρ² · y_{t−2} + sinθ · u_t.

$$H(z) = \sum_{\tau \geq 0} \rho^{\tau} \sin((\tau+1)\theta) z^{-\tau} = \frac{\sin\theta}{1 - 2\rho\cos\theta\, z^{-1} + \rho^2 z^{-2}}$$

Proof sketch. Evaluate the generating function as the imaginary part of e^{iθ}/(1 − ρe^{iθ}z^{−1}); the numerator reduces to

sinθ. The denominator is exactly the characteristic polynomial of the two-tap recursion. Full derivation: Appendix A.1.

Remark (withdrawn claim). This is real-field equivalence of one LTI operator, not bitwise identity. The FFT path and the recurrence have different rounding orders, so unconditional fp32 bitwise equality is false. The earlier claim of unconditional bitwise agreement is withdrawn; the streaming path is bitwise on its own past and the parallel path agrees within tolerance.

Theorem 2 (Strict causality).
For every t, y_t is a function of u_{1:t} only. *Proof sketch.* The recursion reads y_{t−1}, y_{t−2} and u_t; induction on t gives the claim. Perturbing any u_s with s > t leaves y_{1:t} unchanged (Appendix A.2).

Theorem 3 (Constant state).
There exists C, depending only on the architecture width, the number of basis functions and the numeric format, such that the streaming state occupies C bytes for every L. *Proof sketch.* The recurrence stores a fixed number of scalar registers per channel; no term in the update is indexed by absolute position (Appendix A.3).

Theorem 4 (Toeplitz ⇔ time-invariant parameters).
Consider y_t = a_t y_{t−1} + b_t y_{t−2} + c_t u_t with c_t ≠ 0 for all t. The impulse response h(t,s) depends only on t − s if and only if (a_t, b_t, c_t) is constant in t.

$$h(s,s) = c_s, \quad h(s+1,s) = a_{s+1}c_s, \quad h(s+2,s) = (a_{s+2}a_{s+1} + b_{s+2})c_s$$

Proof sketch. Toeplitz at delay 0 forces c_t ≡ c; at delay 1, a_{s+1}c ≡ κ_1 forces a_t ≡ a; at delay 2, (a² + b_{s+2})c ≡ κ_2 forces b_t ≡ b. With a = 2ρcosθ, b = −ρ², c = sinθ and θ in (0,π), constant (a,b,c) implies constant (ρ,θ). The converse is Theorem 1. Full induction: Appendix A.4.

Corollary (duality loss). If ρ_t or θ_t varies with t, no single frequency-domain filter represents the operator, so the single-FFT convolution path does not apply. This concerns that specific path; it does not assert the nonexistence of every O(L log L) algorithm. Earlier statements describing mutual exclusivity between *content addressing* and duality are withdrawn (Section 4).

Theorem 5 (Impossibility for fixed linear summaries).
Assume (i) the history enters the final computation only through a fixed-dimensional linear summary A = Σ_p [α_p k_p + β_p v_p] with A independent of q, (ii) q enters only through the current position, (iii) values may be assigned independently, and (iv) at least two distinguishable targets exist. Then no read-out f(A, q), linear or nonlinear, can implement content addressing.

Proof sketch. Fix the keys. A is a linear map from (v_1,…,v_n) in R^{nd} to R^d. For n ≥ 2, d < nd, so a nonzero Δv lies in its kernel; Δv can be chosen to change the target value. Two distinct value assignments then give identical A and identical q, hence identical f(A,q), but require different answers — a contradiction. The linear-read-out version follows by subtracting two queries. Full proof: Appendix A.5.

Corollary B1.1 (explanatory framework; NOT a theorem). Escaping this boundary requires a term of the form Σ_p φ(q), ψ(k_p) v_p. This is a necessary-condition heuristic supported by our ablations, not a proved characterization of all possible mechanisms; other non-pairwise mechanisms are not excluded.

Scope of Theorem 5. Three constructions leave its premise: deep stacks with per-position nonlinearities that mix A and q; input-dependent kernels (which instead fall under Theorem 4); and outer-product states Σ_p k_p v_p^T that preserve binding. Extension to deep nonlinear stacks is not proved here.

Theorem 6 (Information-theoretic boundary).
Let the state take at most |S| ≤ 2^{8B} distinct values. (a) Lossless reconstruction of x_{1:L} for unbounded L is impossible, since |X|^L eventually exceeds |S|. (b) If the task output f(x) lies in a finite set F with |F| ≤ |S|, encoding f(x) directly in the state is lossless. Hence the assertion that a constant state must lose is false in general; the correct statement is that the *spectral compressed state* used by SGC loses on content addressing. Proof: Appendix A.6.

Two conditional propositions.
Proposition 7 (perturbation bound). Under a floating-point model with a uniform injection bound δ_max, |h(τ)| ≤ ρ^τ/|sinθ| gives |e| ≤ δ_max / ((1 − ρ)|sinθ|). This is conditional on the uniform-bound model; the un-normalized variant is not a valid bound. Appendix A.7.

Proposition 8 (asymptotic rank deficiency). As θ_m → 0, sin((τ+1)θ_m) = (τ+1)θ_m + O(θ_m³), so the dictionary columns become collinear in ρ^τ(τ+1) and the Gram matrix becomes asymptotically singular. The direction is analytic; all reported condition numbers are measured values, not theorems. Appendix A.8.

3.7 Proofs

Complete proofs of Theorems 1–6. Propositions 7–8 are derived in Appendix A.

Proof of Theorem 1
Let ζ = ρe^{iθ}. Then Σ_{τ≥0} ζ^τ z^{−τ} = 1/(1 − ζz^{−1}). Since sin((τ+1)θ) = Im e^{i(τ+1)θ}, the generating function equals Im[e^{iθ}/(1 − ρe^{iθ}z^{−1})]. Writing the denominator as 1 − ρcosθ z^{−1} − iρsinθ z^{−1} and rationalizing gives numerator sinθ(1 − ρcosθ z^{−1}) + ρ sinθ cosθ z^{−1} = sinθ and denominator 1 − 2ρcosθ z^{−1} + ρ²z^{−2}. Hence y_t − 2ρcosθ y_{t−1} + ρ²y_{t−2} = sinθ u_t. Both paths therefore compute the same rational transfer function H(z) over R. Floating-point evaluation order differs, so equality is up to tolerance on the FFT path; the streaming path is bitwise exact with respect to its own past perturbations.

Proof of Theorem 2
Base case: y_1 = sinθ u_1 (with y_0 = y_{−1} = 0) depends only on u_1. Induction: y_t depends on y_{t−1}, y_{t−2}, u_t, all functions of u_{1:t}. Hence y_t is independent of u_s for s > t. Equivalently, adding any perturbation to u_s with s > t changes y_k only for k ≥ s; the prefix y_{1:t} is unchanged bit-for-bit.

Proof of Theorem 3
The streaming update stores, for each of D channels and each of M basis functions, two previous outputs, plus the gate registers. With K scalar registers per channel the state is 4(2MD + (K−1)D) bytes in fp32, an expression with no dependence on L. The specific byte count is an observation; the L-independence is the theorem.

Proof of Theorem 4

Necessity (Toeplitz implies constant). The impulse responses at delays 0, 1 and 2 are h(s,s) = c_s, h(s+1,s) = a_{s+1}c_s and h(s+2,s) = (a_{s+2}a_{s+1} + b_{s+2})c_s. Toeplitz requires each to be independent of s. From delay 0, c_s ≡ c. The architecture's θ clamp guarantees c = sinθ ≠ 0. Delay 1 then gives a_{s+1} ≡ κ_1/c, so a_t ≡ a. Delay 2 gives b_{s+2} ≡ κ_2/c − a², so b_t ≡ b. With a = 2ρcosθ, b = −ρ², c = sinθ, we obtain ρ² constant, hence ρ constant (ρ > 0), then cosθ = a/(2ρ) constant, hence θ constant on (0,π). Sufficiency (constant implies Toeplitz) is Theorem 1. Therefore a non-Toeplitz impulse response is equivalent to some parameter varying with t, and no single frequency-domain filter exists. This negates the single-FFT convolution path; it does not negate every possible sub-quadratic algorithm.

Proof of Theorem 5

Linear read-out. With W linear, y_L = WA + K(0)WE(q). For two queries pointing to $j_1 \neq j_2$, correctness requires WA + K(0)WE(q_1) = E(v_{j_1}) and WA + K(0)WE(q_2) = E(v_{j_2}). Subtracting removes A and yields K(0)W(E(q_1) − E(q_2)) = E(v_{j_1}) − E(v_{j_2}). The left side is independent of the value assignment; the right side varies with it. Contradiction.

Arbitrary read-out. Fix keys and positions. The summary A = Σ_p [α_p k_p + β_p v_p] + const is a linear map L: R^{nd} → R^d. For n ≥ 2, dim ker L ≥ nd − d > 0, so there exists Δv ≠ 0 with L(Δv) = 0. Choose Δv supported on a coordinate that changes the target value v_{j(q)}; this is possible because values are independently assignable and at least two distinguishable targets exist. Let assignments v and v + Δv define two inputs. They share the same history summary A, the same query q, and hence the same output f(A,q), but the required answers differ. No read-out f, linear or nonlinear, can satisfy both.

Premises explicitly used: q enters only via the current position; coefficients (α_p, β_p) are content-independent; the summary dimension is fixed in n; values are independent; at least two targets are distinguishable. Key linear independence is not required. The proof does not cover deep stacks with per-position nonlinearities, input-dependent kernels, or outer-product states (Section 4).

Proof of Theorem 6

(a) A state with B bytes has |S| ≤ 2^{8B} configurations. Lossless reconstruction requires the map x_{1:L} s_L to be injective, hence |X|^L ≤ |S| for every L. Since |X| > 1 and L is unbounded, this fails for large L by the pigeonhole principle. (b) If f(x) in F and |F| ≤ |S|, let the state encode f(x) exactly; the required output is then recoverable with no loss. Therefore the impossibility in (a) concerns reconstruction of unbounded histories, not the constant-state paradigm as such.

4 Scope Correction: Tensor Spectral Convolution

SGC constrains its *kernel* to be a static LTI operator; it does not constrain the tensor order of its *input signal* . Replacing the vector signal by an outer product and the read-out by a contraction yields Tensor Spectral Convolution (TSC):

$$u_t[r,n] = a_t[r]\, b_t[n], \quad a_t = \mathrm{Code}[x_{t-1}], \quad b_t = W_V E(x_t)$$

$$s_{p,r,n,t} = \rho_{p,n} s_{p,r,n,t-1} + u_t[r,n], \quad z_n = \sum_{p,r} w_{p,n}\, q_r\, s_{p,r,n,L}$$

Because ρ_{p,n} is independent of t and r, the impulse response depends only on the delay τ; the kernel matrix is Toeplitz channel-wise, so Theorems 1–3 remain valid. The contraction Σ_r q_r s_{p,r,n} implements the (q · k_τ) match outside the recurrence. Theorem 5 does not apply (tensor signal; bilinear read-out) and Theorem 4 is not triggered (no time-varying parameters). Consequently **content addressing and FFT/IIR duality are not mutually exclusive** ; the mutual exclusivity is between time-varying parameters and duality. This corrects the earlier claim.

The key dictionary is a fixed orthogonal code with Code^{T}Code = (V/R)·I, constructed deterministically (Sylvester Hadamard when V = 2^k). It is not learned and does not use labels; it is the signal-side analogue of SGC's fixed (ρ,θ) basis. For ρ → 1 and one pole, the state Σ_τ k_τ v_τ^T is isomorphic to a linear-attention state. We record this isomorphism openly: the construction has no pairwise scoring matrix, no softmax normalization and no growing KV cache, and its ρ < 1 decay spectrum has no attention counterpart, but whether it belongs to the linear-attention family is left to the reader.

Cost. The state holds P·R·N scalars, so O(1) is asymptotic with a substantial constant (56,320 B in our configuration versus 512 B for the original mixer). A coarse analytic crossover suggests no compute advantage below L ≈ 880; this has not been measured against an attention implementation.

5 Finite-Sample Observations (Not Theorems)

Each item below is a finite-sample estimate under a fixed configuration; none is a theorem and none claims universality. Precision, seed count and provenance are in Appendix D.

O1 — saturated transfer ceiling.

In a 1.7B-parameter weight-transfer fitting study, the 28-layer residual-stream cosine reached 0.2484 against a 0.2476 static-kernel oracle, while a same-input single-layer control reached 0.5430 (global-average kernel 0.4478). All alignment metrics improved across four versions while generation remained non-sentential. The oracle is a finite-layer estimator, not a proved upper bound; 0.2484 > 0.2476 is consistent with that reading.

O2 — phase-resonance content addressing.

A fixed-frequency-slot variant raised single-hop accuracy from 0.4160 to 0.8672, whereas an equal-parameter low-rank attention baseline reached 1.0000 (5,217 vs 5,152 parameters).

O3 — native pilot training.

A 470K-parameter native model trained on 3.07M tokens of repository text reached validation perplexity 10.817 against a same-budget Transformer's 11.281, and 10.217 with output-distribution distillation. Generation was non-sentential on all three arms. Perplexity and residual alignment are not sufficient indicators of usable language ability.

O4 — non-replication of the periodic-axis gain.

An external white-box run reported a periodic-axis advantage (0.9844 vs 0.8535). A subsequent in-repository probe with seven grids produced 0.1094–0.1354, all below the copy-last baseline of 0.1575, with a paired confidence interval crossing zero. The advantage is therefore not treated as established.

O5 — parity and parameter efficiency of TSC.

On a single-hop synthetic binding task (V = 64, L = 9, held-out

512 per seed), the tensor-signal model W1 reached 1.0000 over 5 seeds (5×512 = 2560 held-out items) at 10264 parameters, against 0.9987 for a historical windowed softmax-attention record at 10240 parameters (Fisher exact p = 0.4998, statistically indistinguishable). A reduced arm W4_half reached 1.0000 with 4920 parameters (48.0% of the reference) over 5 seeds; a further arm M5 reached 1.0000 with 2792 parameters (3.67× efficiency) over 5 seeds. Trained at L = 17, an analytic evaluation identical sample-by-sample to the forward pass stayed at 1.0000 up to L = 1025 (10 lengths tested). A value-permutation control stayed at 1.0000, while removing the predecessor-key encoding collapsed accuracy to 0.2331. A random orthogonal code also reached 1.0000, so the result does not depend on the Hadamard construction specifically. A two-hop variant collapsed to 0.0322 (chance level), so single-layer contraction supports one hop only.

O6 — physical Pareto frontier.
With the state register cast at every step, fp64 and fp32 stayed at 1.0000 through L = 4097; bf16 and int8 first dropped at L = 2049 (0.5312); fp16 dropped at L = 4097 (0.6250). A capacity probe with 128 pairs required at least 1,280 bits of information against a 225,280-bit fp16 state (176× headroom), and input noise up to 0.20 left accuracy unchanged, so Shannon capacity is not the limiting factor. The decay coefficient is the only effective lever: fp32 with ρ = 0.999 (BIBO-stable) stayed at 1.0000 to L = 16,385. Retraining from scratch with ρ frozen at 0.999 gave 1.0000 on 5 seeds with zero between-seed variance, and 1.0000 in all four tested precisions on held-out data. Same-sample prefixes and fixed-scale quantization controls showed that dynamic scaling is an effective adaptation rather than a defect and that the fp16 non-monotonicity is a genuine numerical effect. Measured reference-implementation cost stayed near 35.25 µs/token with a constant 56320 B state. Conservation checks passed 43/43.

6 Limitations and Missing Evidence

- No language-modeling main experiment at ≥10M tokens with equal-compute Mamba-2, GLA, DeltaNet, Hyena and Transformer baselines has been run. O3 is a 3.07M-token pilot only.
- Not every main result has a paired confidence interval across ≥5 seeds. O5 and O6 do; the transfer, phase-resonance, pilot and non-replication observations do not.
- All positive TSC results are on a synthetic single-hop task with a 64-token vocabulary and a single layer. Length extrapolation beyond L = 129 permits repeated keys (with a unique query key), so it is an extrapolation test, not an in-distribution test.
- Multi-hop retrieval fails at chance level in the present single-layer construction; stacking and recurrence have not been evaluated.
- int8 quantization remains a hard boundary at long lengths; fixed-scale, per-channel and dynamic schemes were all measured, none removed it. ρ = 1 is only critically stable and divergent for unbounded streams, so it is not recommended.
- Theorem 5 is not extended to deep stacks with per-position nonlinearities; Corollary B1.1 is an explanatory framework, not a proved necessity.
- The parity result in O5 compares against a *windowed* attention record read from a stored result file; a full global attention baseline and a measured throughput comparison are absent. Parity on one synthetic task is not general-capability parity.
- The reported throughput is from an unoptimized reference implementation; only its scaling with L is informative, not its absolute rate.

7 Related Work

Structured state-space and long-convolution models (arXiv:2111.00396, 2208.04933, 2302.10866) established the LTI convolution/recurrence duality and noted that a convolution view requires time invariance. Selective and linear-attention state models (arXiv:2312.00752, 2405.21060, 2312.06635, 2406.06484, 2006.16236, 2305.13048, 2407.04620) obtain content addressing through input-dependent state updates at the cost of the LTI convolution view. Hybrid and long-context systems (arXiv:2307.02486, 2403.19887) retain attention explicitly. Zoology and related studies (arXiv:2312.04927, 2402.01032, 2404.08819) already established that fixed-state models are weaker on recall and copying, including TC^0-class arguments.

We do not claim to be the first to establish any of those boundaries. Our additions are the algebraic reduction to checkable properties of the SGC operator (Theorems 1–6), the equivalence strengthening of Theorem 4, the arbitrary-read-out version of Theorem 5, and the constructive scope correction of Section 4. The comparison above is qualitative and bibliographic; equal-parameter and equal-compute quantitative comparisons have not been performed.

8 Conclusion

Six theorems delimit what a constant-state spectral convolution can express: exact duality, strict causality and constant state hold; time-varying parameters are exactly the obstruction to a Toeplitz kernel; a fixed linear history summary forbids content addressing for any read-out; and a finite state is lossy only for unbounded reconstruction, not for finite-valued tasks. Tensor signals with a contraction read-out evade the third boundary without sacrificing the first, so duality and content addressing coexist. Measured results are reported strictly as finite-sample observations. Scaling these findings to language modeling, multi-hop retrieval and equal-compute baselines remains open work, not a conclusion of this paper.

References

Entries are keyed by arXiv identifier and verified title/venue to avoid transcription errors in author metadata.

[1] arXiv:2111.00396 — Efficiently Modeling Long Sequences with Structured State Spaces. ICLR 2022.

[2] arXiv:2208.04933 — Simplified State Space Layers for Sequence Modeling. 2022.

[3] arXiv:2302.10866 — Hyena Hierarchy: Towards Larger Convolutional Language Models. 2023.

[4] arXiv:2312.00752 — Mamba: Linear-Time Sequence Modeling with Selective State Spaces. COLM 2024.

[5] arXiv:2405.21060 — Transformers are SSMs: Generalized Models and Efficient Algorithms Through Structured State Space Duality. ICML 2024.

[6] arXiv:2312.06635 — Gated Linear Attention Transformers with Hardware-Efficient Training. 2023.

[7] arXiv:2406.06484 — Parallelizing Linear Transformers with the Delta Rule over Sequence Length. 2024.

[8] arXiv:2006.16236 — Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. ICML 2020.

[9] arXiv:2305.13048 — RWKV: Reinventing RNNs for the Transformer Era. EMNLP 2023.

[10] arXiv:2407.04620 — Learning to (Learn at Test Time): RNNs with Expressive Hidden States. 2024.

[11] arXiv:2307.02486 — LongNet: Scaling Transformers to 1,000,000,000 Tokens. 2023.

[12] arXiv:2403.19887 — Jamba: A Hybrid Transformer-Mamba Language Model. 2024.

[13] arXiv:2312.04927 — Zoology: Measuring and Improving Recall in Efficient Language Models. ICLR 2024.

[14] arXiv:2402.01032 — Repeat After Me: Transformers are Better than State Space Models at Copying. 2024.

[15] arXiv:2404.08819 — The Illusion of State in State-Space Models. ICML 2024.

Broader Impact

This work is theoretical and empirical on synthetic tasks. It does not involve human subjects, personal data, deployed systems, or dual-use capabilities. Its positive potential is a clearer account of when constant-state sequence models are sufficient, which can reduce the memory and energy cost of long-context inference and help practitioners avoid over-engineering attention-free architectures for tasks that structurally require content addressing. Its main risk is interpretive: a reader may take the measured parity on a single synthetic binding task as evidence of general language-model parity. We have therefore stated the scope of every observation, listed the missing large-scale experiments explicitly, and declined to make a superiority claim. A second risk is that negative results are used to dismiss an entire family of architectures; Theorem 6(b) and Section 4 specifically argue against that reading. We also disclose that generative AI tools were used during implementation and language editing; all research questions, conjectures and conclusions were specified by the human author, who takes full responsibility for the content.

Appendix A Supplementary Derivations

The full proofs of Theorems 1–6 appear in Section 3.7. This appendix collects the two conditional propositions and the numerical conventions used in the observations.

A.7 Proposition 7

The impulse response of the two-tap system has magnitude bounded by $\rho^{\tau}/|\sin\theta|$, because the poles are $\rho e^{\pm i\theta}$. A per-step additive perturbation bounded by δ_{max} accumulates through the same stable filter, giving total error at most $\delta_{max} \Sigma_{\tau} \rho^{\tau}/|\sin\theta| = \delta_{max}/((1-\rho)|\sin\theta|)$. The bound is conditional on a uniform injection bound; using an un-normalized u as δ is not a valid bound.

A.8 Proposition 8

Taylor expansion gives $\sin((\tau+1)\theta_m) = (\tau+1)\theta_m + O(\theta_m^3)$. Hence $\psi_m(\tau) \approx \theta_m \rho^{\tau}(\tau+1)$, so as $\theta_m \to 0$ all such columns become proportional to the same vector $\rho^{\tau}(\tau+1)$ and the Gram matrix loses rank asymptotically. This proves the direction of degeneracy only. Any numerical condition number is a measured quantity.

Appendix B Conjectures and Open Questions

The following are explicitly unverified. None is used as a premise in any theorem or as a claimed result.

· **C1.** Stacking TSC layers or adding inter-layer recurrence restores multi-hop retrieval. Motivated by the two-hop collapse at chance level in O5; no stacking experiment exists.

· **C2.** A deterministic orthogonal code scales to realistic vocabularies V 64. Hadamard requires V = 2^k; deterministic random orthogonal projections are untested beyond V = 256.

· **C3.** Theorem 5 extends to deep stacks with per-position nonlinearities. Per-position nonlinearities can produce cross-terms between history tokens, so the fixed-summary argument does not directly transfer.

· **C4.** Freezing ρ = 0.999 from the start of training remains stable on language modeling at ≥10M tokens. Verified only on a synthetic single-hop task with five seeds.

· **C5.** Layered quantization or residual accumulation pushes the int8 long-length boundary further. Fixed scalar and per-channel scales were measured and were worse than dynamic scaling; other schemes remain untried.

· **C6.** On equal compute, the periodic/continuous-signal axis and the content-addressing axis are complementary, motivating a small-attention hybrid. The two-axis evidence comes from synthetic tasks at unequal budgets; the periodic-axis gain was not replicated (O4).

· **C7.** TSC with a contraction read-out is not equivalent to any linear-attention family member once ρ < 1 and multiple poles are retained. The ρ → 1, single-pole limit is isomorphic; the general case is open.

· **C8.** The decay spectrum provides an inductive bias on oscillatory and damped physical signals that attention lacks. Plausible from the basis definition; no measured comparison exists.

Appendix C Falsified Conjectures

These were proposed, implemented where feasible, and rejected on evidence. They are retained because they define the boundary of the search.

· **F1.** Extending the memory window by increasing ρ_max improves retrieval. Rejected: measured accuracy did not improve and decreased at ρ_max = 0.9999 on a 65-token sequence already covered by the shorter window.

· **F2.** A state-feedback modulation of ρ_t and θ_t (SFSC) preserves FFT duality. Rejected: the resulting LTV impulse response is non-Toeplitz; FFT-proxy relative error 0.9720. This is also the empirical instance behind Theorem 4.

· **F3.** Phase-resonance frequency slots exceed equal-parameter attention. Rejected: 0.8672 against 1.0000 at comparable parameter counts (O2).

· **F4.** Content addressing and FFT/IIR duality are mutually exclusive. Rejected by the TSC construction in Section 4; the correct exclusivity is between time-varying parameters and duality.

· **F5.** An oscillatory basis itself provides the required inductive advantage. Rejected: a random orthogonal basis reached 1.0000 on the relevant task, while a damped-oscillator grid did not; condition number and orthogonality, not the sinusoidal form, accounted for the measured difference.

· **F6.** Dynamic quantization scaling causes spurious non-monotonicity. Rejected by same-sample prefix controls: the fp16 non-monotonicity persists, and fixed-scale quantization degrades earlier.

· **F7.** Shannon capacity limits long-length retrieval in the measured configuration. Rejected: 176× headroom and noise tolerance up to 0.20 isolate the decay coefficient as the operative limit.

Appendix D Observation Provenance

All values below were computed from the raw result files accompanying the anonymized supplementary artifact. No value was transcribed from a secondary summary.

Obs.	Config / seeds	Key values	File
O1	1.7B transfer, 1 run	0.2484 / 0.2476 / 0.5430	transfer study
O2	single-hop, 3 seeds	0.8672 vs 1.0000	phase-resonance probe
O3	470K, 3.07M tokens	10.817 / 11.281 / 10.217	native pilot
O4	7 grids, 1 run	0.1094–0.1354 vs 0.1575	periodic-axis probe
O5	V=64, 5 seeds, 512/seed	1.0000 / 0.9987 / p=0.4998	TSC probe
O6	5 seeds, 4 precisions	L=16385, 43/43 checks	physics probes

Measured configuration details. O5: W1 has 10264 parameters, W4_half 4920, M5 2792; the reference record has 10240. The paired bootstrap on the general second-order dictionary gave a mean difference of 0.1901 with 95% CI [0.1582, 0.2220] over 3 seeds (sample-internal variance only). O6: the frozen-ρ retraining gave held-out means [1.0, 1.0, 1.0, 1.0] with between-seed standard deviations [0.0, 0.0, 0.0, 0.0]; L = 16,385 accuracy mean 1.0000, minimum 1.0000; per-token cost at L = 4097 was 35.25 µs with a 56320 B state. Conservation guards on the final ρ reported FFT and IIR relative errors of order 1e−16.

Appendix E Reproducibility and Tool Use

Every theorem is derived analytically in Appendix A and requires no execution. Every observation is produced by a deterministic script with recorded seeds, precision and sample counts; the raw result files, gradient finite-difference checks, conservation assertions and consistency checks are included in the anonymized supplementary artifact. Gradient checks passed 8/8, 10/10, 28/28 and 44/44 across the four probe generations; the physical consistency checker passed 43/43. Whole-sequence forward evaluation and the analytic evaluator agree sample by sample.

Generative AI tools were used for implementation assistance, experiment scripting and language editing. They did not originate the research questions, the architectural conjectures or the conclusions, and no experimental data, citation or result was generated by such a tool. The human author verified every numerical claim against the raw files and accepts full responsibility for the manuscript.

Appendix F Withdrawn Claims Register

The following statements appeared in earlier working versions and are formally withdrawn. They are listed so that the revision is auditable.

Withdrawn statement	Reason
Unconditional fp32 bitwise duality	Different rounding orders; Theorem 1 is real-field only
Content addressing ↔ duality exclusive	TSC is a counterexample (Section 4)
Periodic-axis superiority over attention	Not replicated in-repository (O4)
Equivalent 4–7B parameter claim	No supporting measurement
B1 holds for any read-out unrestricted	Premise must be fixed linear summary; Thm 5 states it
Dynamic scaling causes spurious effects	Same-sample prefix controls (O6)